Python Programming - 2301CS404

Lab - 7

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01) WAP to find sum of tuple elements.

Sample Input:

```
T = (10, 20, 30, 40)
```

Sample Output:

```
Sum of tuple elements = 100
```

In [1]:
```python
T = tuple(map(int,input("Enter Elements:").split()))
sum=0
for i in range(len(T)):
    sum = sum + T[i]
print("Sum of Tuple Elements:",sum)
```

```
Sum of Tuple Elements: 100
```

02) WAP to find Maximum and Minimum K elements in a given tuple.

Sample Input:

```
T = (7, 2, 9, 4, 1, 5)
K = 2
```

Sample Output:

```
Minimum 2 elements: (1, 2)
Maximum 2 elements: (9, 7)
```

In [3]:
```
T = tuple(map(int,input("Enter Number").split()))
K = int(input("Enter k:"))

lst = list(T)
lst.sort()

min_k = lst[:K]
max_k = lst[-K:]

print("Minimum K elements:", min_k)
print("Maximum K elements:", max_k)
```

```
Minimum K elements: [1, 2]
Maximum K elements: [7, 9]
```

03) WAP to find tuples which have all elements divisible by K from a list of tuples.

Sample Input:

```
tuple_list = [(2, 4, 6), (3, 6, 9), (4, 8, 12), (5, 10, 15)]
K = 2
```

Sample Output:

```
Tuples divisible by 2 : [(2, 4, 6), (4, 8, 12)]
```

In [4]:
```
lst = [(2, 4, 6), (3, 6, 9), (2, 3, 4), (8, 10, 12)]
K = 2

result = []

for t in lst:
    count = 0

    for num in t:
        if num % K == 0:
            count += 1

    if count == len(t):
        result.append(t)

print("Tuples divisible by 2:", result)
```

```
Tuples divisible by 2: [(2, 4, 6), (8, 10, 12)]
```

04) WAP to create a list of tuples from given list having number and its cube in each tuple.

Sample Input:

```
numbers = [1, 2, 3, 4, 5]
```

Sample Output:

```
List of tuples with number and its cube: [(1, 1), (2, 8), (3,
27), (4, 64), (5, 125)]
```

In [6]:
```python
numbers = [1, 2, 3, 4, 5]

res = []

for i in numbers:
    cube = i ** 3
    res.append((i, cube))

print("List of tuples with number and its cube:", res)
```

```
List of tuples with number and its cube: [(1, 1), (2, 8), (3, 27), (4, 64), (5, 1
25)]
```

05) WAP to find tuples with all positive elements from the given list of tuples.

Sample Input:

```
tuple_list = [(1, 2, 3), (-1, 2, 3), (4, 5, 6), (0, 1, 2), (-3,
-4, -5)]
```

Sample Output:

```
Tuples with all positive elements: [(1, 2, 3), (4, 5, 6)]
```

In [14]:
```python
tuple_list = [(1, 2, 3), (-1, 2, 3), (4, 5, 6), (0, 1, 2), (-3, -4, -5)]
res=[]
for t in tuple_list:
    count = 0
    for num in t:
        if num>0:
            count+=1

    if count == len(t):
        res.append(t)

print("Tuples with all positive elements: ",res)
```

```
Tuples with all positive elements:  [(1, 2, 3), (4, 5, 6)]
```

06) WAP to remove tuples of length K.

Sample Input:

```
tuple_list = [(1, 2), (3, 4, 5), (6,), (7, 8, 9), (10, 11)]
K = 2
```

```
:**Sample Output:**
```

List after removing tuples of length 2 : [(3, 4, 5), (6,), (7, 8, 9)]

In [18]:
```
tuple_list = [(1, 2), (3, 4, 5), (6,), (7, 8, 9), (10, 11)]
K=2
res=[]
for t in tuple_list:
    if len(t) != K:
        res.append(t)
print("List after removing tuples of length 2:",res)
```

```
List after removing tuples of length 2: [(3, 4, 5), (6,), (7, 8, 9)]
```

07) WAP to remove all occurrences of a given element from a tuple.

Sample Input:

```
T = (3, 5, 3, 7, 3, 9)
x = 3
```

Sample Output:

```
(5, 7, 9)
```

In [27]:
```
T =(3,5,3,7,3,9)
x=3
li = list(T)

for x in li:
    li.remove(x)

print(tuple(li))
```

```
(5, 7, 9)
```

08) WAP to remove duplicates from tuple.

Sample Input:

```
T = (3, 5, 3, 7, 3, 9)
```

Sample Output:

```
(3, 5, 7, 9)
```

In [28]:
```
T = (3, 5, 3, 7, 3, 9)
res=[]
for num in T:
    if num not in res:
      res.append(num)
```

```
print(tuple(res))
```

```
(3, 5, 7, 9)
```

09) WAP to multiply adjacent elements of a tuple and print that resultant tuple.

Sample Input:

```
t = (2, 3, 4, 5)
```

Sample Output:

```
Resultant tuple after multiplying adjacent elements: (6, 12, 20)
```

In [30]:
```
t = (2, 3, 4, 5)
res=[]
for i in range(len(t)-1):
    res.append(t[i]*t[i+1])

print("Resultant tuple after multiplying adjacent elements:",tuple(res))
```

```
Resultant tuple after multiplying adjacent elements: (6, 12, 20)
```

10) WAP to test if the given tuple is distinct or not.

Sample Input:

```
tuple1 = (1, 2, 3, 4)
tuple2 = (1, 2, 2, 3)
```

Sample Output:

```
(1, 2, 3, 4) is distinct? -> True
(1, 2, 2, 3) is distinct? -> False
```

In [1]:
```
T1 = tuple(map(int,input("Enter Elements:").split()))
T2 = tuple(map(int,input("Enter Elements:").split()))

l1 = list(T1)
l2 = list(T2)
s1 = list(set(T1))
s2 = list(set(T2))

if l1 == s1:
    print(f"{T1} is Distinct->True")
else:
    print(f"{T2} is not Distinct ->False")

if l2 == s2:
    print(f"{T1} is Distinct->True")
else:
    print(f"{T2} is not Distinct ->False")
```

```
(1, 2, 3, 4) is Distinct->True
(1, 2, 2, 3) is not Distinct ->False
```

11) WAP to rotate the elements of a tuple to the right by `k` positions.

Sample Input:

```
T = (10, 20, 30, 40, 50)
k = 2
```

Sample Output:

```
(40, 50, 10, 20, 30)
```

In [5]:
```
T = tuple(map(int,input("Enter Elements:").split()))
K = 2
a = list(T)

rotatedList = a[-K:] + a[:-K]
print(tuple(rotatedList))
```

```
(40, 50, 10, 20, 30)
```

12) WAP to merge two tuples of equal length alternately.

Sample Input:

```
T1 = (1, 3, 5)
T2 = (2, 4, 6)
```

Sample Output:

```
(1, 2, 3, 4, 5, 6)
```

In [30]:
```
T1 = (1, 3, 5)
T2 = (2, 4, 6)

t1 = list(T1)
t2 = list(T2)

t1.extend(t2)
t1.sort()
print(tuple(t1))
```

```
(1, 2, 3, 4, 5, 6)
```

13) WAP to create a tuple of squares of elements that are even and greater than 10.

Sample Input:

```
T = (4, 12, 7, 18, 10)
```

Sample Output:

```
(144, 324)
```

In [36]:
```
T = tuple(map(int,input("Enter Elements:").split()))
l = list(T)

l1 = [i**2 for i in l if i%2==0 and i>10]
print(tuple(l1))
```

```
(144, 324)
```

14) WAP to create (index, value) pairs for prime numbers.

Sample Input:

```
T = (4, 7, 9, 11, 15)
```

Sample Output:

```
((1, 7), (3, 11))
```

In [31]:
```
T = (4, 7, 9, 11, 15)

result = []

for i in range(len(T)):
    num = T[i]

    if num > 1:
        for j in range(2, num):
            if num % j == 0:
                break
        else:
            result.append((i, num))

print(tuple(result))
```

```
((1, 7), (3, 11))
```

15) WAP to split a tuple into even-index and odd-index tuples.

Sample Input:

```
T = ('a', 'b', 'c', 'd', 'e')
```

Sample Input:

```
even = ('a', 'c', 'e')
odd = ('b', 'd')
```

In [2]:
```
T = ('a', 'b', 'c', 'd', 'e')

even = []
```

```
odd = []

for i in range(len(T)):
    if i % 2 == 0:
        even.append(T[i])
    else:
        odd.append(T[i])

even_index = tuple(even)
odd_index = tuple(odd)

print("Even index tuple:", even_index)
print("Odd index tuple:", odd_index)
```

```
Even index tuple: ('a', 'c', 'e')
Odd index tuple: ('b', 'd')
```

16) WAP to compute column-wise sum from a tuple of lists.

Sample Input:

```
T = ([1, 2, 3], [4, 5, 6], [7, 8, 9])
```

Sample Output:

```
12 15 18
```

In [3]:

```
t = ([1, 2, 3],
     [4, 5, 6],
     [7, 8, 9])

result = []

for col in zip(*t):
    result.append(sum(col))

print("Column-wise sum:", result)
```

```
Column-wise sum: [12, 15, 18]
```

In [ ]:

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In [ ]: